

Assignment: Project Search Statistical Initial Results

Run your agent on as many instances of your environment as possible. Collect statistics on the agent's performance. If multiple options are available for your problem (heuristics, evaluation functions, etc.) gather statistics on all of them for comparison.

Note that this is also a good chance to fix bugs and improve agent performance.

Store the PDF document in the `project` folder of the repository as well.

Required Submissions

- A PDF with a table of your statistical results, via Canvas.

Explanation of Statistical Results:

"Maze" is the name of my environment. The following statistics below are the result of the Maze environment being used by my agent1 which in the table below showcases the data of five episodes and their associated rewards and steps taken to reach the end of the maze. Currently there are three difficulties associated with differing dimensions of the maze and as the table shows agent1 was tested on all available difficulties. To give more meaning to the rewards received from agent1 I also provided similar tests ran on my random agent, i.e., the closer to zero the reward is, the better.

agent1

Easy (7x7)	Medium (14x14)	Hard (28x28)
Episode 1: Reward = 46, Steps = 36 Episode 2: Reward = 48, Steps = 36 Episode 3: Reward = 90, Steps = 16 Episode 4: Reward = 65, Steps = 30 Episode 5: Reward = 79, Steps = 24 Results over 5 episodes: Average reward: 65.60 Average steps: 28.40	Episode 1: Reward = -28, Steps = 54 Episode 2: Reward = -441, Steps = 108 Episode 3: Reward = -148, Steps = 74 Episode 4: Reward = -269, Steps = 90 Episode 5: Reward = -324, Steps = 96 Results over 5 episodes: Average reward: -242.00 Average steps: 84.40	Episode 1: Reward = -5683, Steps = 344 Episode 2: Reward = -2190, Steps = 218 Episode 3: Reward = -6525, Steps = 368 Episode 4: Reward = -2277, Steps = 222 Episode 5: Reward = -965, Steps = 150 Results over 5 episodes: Average reward: -3528.00 Average steps: 260.40
Episode 1: Reward = 92, Steps = 16 Episode 2: Reward = 94, Steps = 14 Episode 3: Reward = 92,	Episode 1: Reward = -71, Steps = 62 Episode 2: Reward = -93, Steps = 66 Episode 3: Reward = -188,	Episode 1: Reward = -2877, Steps = 248 Episode 2: Reward = -965, Steps = 150 Episode 3: Reward = -1861,

Steps = 14 Episode 4: Reward = 79, Steps = 24 Episode 5: Reward = 42, Steps = 38 Results over 5 episodes: Average reward: 79.80 Average steps: 21.20	Steps = 80 Episode 4: Reward = -105, Steps = 68 Episode 5: Reward = -134, Steps = 72 Results over 5 episodes: Average reward: -118.20 Average steps: 69.60	Steps = 202 Episode 4: Reward = -5956, Steps = 352 Episode 5: Reward = -6894, Steps = 378 Results over 5 episodes: Average reward: -3710.60 Average steps: 266.00
Episode 1: Reward = 90, Steps = 16 Episode 2: Reward = 88, Steps = 20 Episode 3: Reward = 92, Steps = 16 Episode 4: Reward = 48, Steps = 36 Episode 5: Reward = 81, Steps = 22 Results over 5 episodes: Average reward: 79.80 Average steps: 22.00	Episode 1: Reward = -134, Steps = 72 Episode 2: Reward = -342, Steps = 98 Episode 3: Reward = -239, Steps = 86 Episode 4: Reward = -506, Steps = 114 Episode 5: Reward = 19, Steps = 44 Results over 5 episodes: Average reward: -240.40 Average steps: 82.80	Episode 1: Reward = -6311, Steps = 362 Episode 2: Reward = -4280, Steps = 300 Episode 3: Reward = -4341, Steps = 302 Episode 4: Reward = -722, Steps = 132 Episode 5: Reward = -572, Steps = 120 Results over 5 episodes: Average reward: -3245.20 Average steps: 243.20

agent_random

Easy (7x7)	Medium (14x14)	Hard (28x28)
Episode 1: Reward = -69156, Steps = 500 Episode 2: Reward = -78569, Steps = 500 Episode 3: Reward = -74472, Steps = 500 Episode 4: Reward = -75935, Steps = 500 Episode 5: Reward = -64423, Steps = 500 Results over 5 episodes: Average reward: -72511.00 Average steps: 500.00	Episode 1: Reward = -63035, Steps = 500 Episode 2: Reward = -74202, Steps = 500 Episode 3: Reward = -67621, Steps = 500 Episode 4: Reward = -68692, Steps = 500 Episode 5: Reward = -57908, Steps = 500 Results over 5 episodes: Average reward: -66291.60 Average steps: 500.00	Episode 1: Reward = -72494, Steps = 500 Episode 2: Reward = -65826, Steps = 500 Episode 3: Reward = -62628, Steps = 500 Episode 4: Reward = -52483, Steps = 500 Episode 5: Reward = -66755, Steps = 500 Results over 5 episodes: Average reward: -64037.20 Average steps: 500.00
Episode 1: Reward = -58629, Steps = 500 Episode 2: Reward = -62809, Steps = 500 Episode 3: Reward = -71527, Steps = 500	Episode 1: Reward = -70577, Steps = 500 Episode 2: Reward = -59402, Steps = 500 Episode 3: Reward = -72147, Steps = 500	Episode 1: Reward = -67192, Steps = 500 Episode 2: Reward = -65106, Steps = 500 Episode 3: Reward = -62448, Steps = 500

Steps = 500 Episode 4: Reward = -61285, Steps = 500 Episode 5: Reward = -75716, Steps = 500 Results over 5 episodes: Average reward: -65993.20 Average steps: 500.00	Steps = 500 Episode 4: Reward = -67335, Steps = 500 Episode 5: Reward = -62272, Steps = 500 Results over 5 episodes: Average reward: -66346.60 Average steps: 500.00	Steps = 500 Episode 4: Reward = -57227, Steps = 500 Episode 5: Reward = -66563, Steps = 500 Results over 5 episodes: Average reward: -63707.20 Average steps: 500.00
Episode 1: Reward = -16717, Steps = 270 Episode 2: Reward = -64120, Steps = 500 Episode 3: Reward = -55587, Steps = 500 Episode 4: Reward = -66766, Steps = 500 Episode 5: Reward = -7051, Steps = 178 Results over 5 episodes: Average reward: -42048.20 Average steps: 389.60	Episode 1: Reward = -73516, Steps = 500 Episode 2: Reward = -68677, Steps = 500 Episode 3: Reward = -64378, Steps = 500 Episode 4: Reward = -72310, Steps = 500 Episode 5: Reward = -69135, Steps = 500 Results over 5 episodes: Average reward: -69603.20 Average steps: 500.00	Episode 1: Reward = -66841, Steps = 500 Episode 2: Reward = -60937, Steps = 500 Episode 3: Reward = -72999, Steps = 500 Episode 4: Reward = -65864, Steps = 500 Episode 5: Reward = -67973, Steps = 500 Results over 5 episodes: Average reward: -66922.80 Average steps: 500.00